

# 01 “ Guides & Plan extraction

Exhaustive extraction of the BobaLink reference deliverable’s top-level guide, the 5 formal docs, README, and the two plan documents. Every functional/non-functional requirement, feature, flow, edge case, config, API endpoint, dependency, build/release step, acceptance criterion, milestone, and backend-integration point is captured verbatim with file + section citation. Boba-specific constraint flags are called out explicitly. Source product is named **BobaLink**, v1.0.0, dated 2026-06-06. NOTE: BobaLink is a *reference/spec deliverable*, not the Boba repo’s own naming “ the Boba implementation will rename/re-target.

## Files read

| # | File                                                                             | Lines       |
|---|----------------------------------------------------------------------------------|-------------|
| 1 | Browser Torrent<br>Extension Guide.md (master<br>guide / index)                  | 240         |
| 2 | Browser Torrent<br>Extension Guide/docs/<br>technical-<br>specification.md       | 1152        |
| 3 | Browser Torrent<br>Extension Guide/docs/<br>api-reference.md                     | 2042        |
| 4 | Browser Torrent<br>Extension Guide/docs/<br>developer-guide.md                   | 1334        |
| 5 | Browser Torrent<br>Extension Guide/docs/<br>installation-guide.md                | 949         |
| 6 | Browser Torrent<br>Extension Guide/docs/<br>user-guide.md                        | 865         |
| 7 | Browser Torrent<br>Extension Guide/<br>README .md (byte-identical to file<br>#1) | 240         |
| 8 | Browser Torrent<br>Extension Guide/plan.md<br>(master execution plan)            | 96          |
| 9 | Browser Torrent<br>Extension Guide/<br>implementation-plan.md                    | 410         |
|   | <b>TOTAL</b>                                                                     | <b>7328</b> |

All 9 files read IN FULL (chunked to EOF). Files #1 and #7 are byte-for-byte identical (the master guide is also the README).

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## A. PROJECT OVERVIEW & KEY METRICS

**Source:** master-guide Â§1â€œProject Overviewâ€œ; tech-spec Â§1 â€œExecutive Summaryâ€œ; README.

- BobaLink = cross-browser WebExtension (**Manifest V3**) that detects magnet links and .torrent file references on any web page (and across tab groups), deduplicates by cryptographic infohash, and transmits them to a **Boba orchestration server** OR directly to a **qBitTorrent WebUI** instance.
- Three core operational problems it solves (tech-spec Â§1): (1) Discovery-to-Download latency (eliminate manual copy/paste â€” reduce to single click); (2) Batch operation efficiency (tab groups as atomic units); (3) Operational resilience (offline-aware queue with exponential backoff so transient failures never lose torrents).
- Does NOT perform any BitTorrent protocol operations itself â€” it is a â€œthin clientâ€œ / â€œintelligent scraper and forwarding agentâ€œ (tech-spec Â§2).
- Built on **WXT framework** + **TypeScript**; targets Chrome (â‰¥88), Firefox (â‰¥109), Opera, Yandex Browser; WCAG 2.1 AA; credentials encrypted at rest via **AES-256-GCM** (Web Crypto API).

**Key Metrics at a Glance (tech-spec Â§1 table):** | Metric | Target | |â€”|â€”| | Extension bundle size (compressed) | â‰‰ 350 KB | | Cold-start service worker | â‰‰ 50 ms | | Magnet link detection latency | â‰‰ 5 ms per link | | API call timeout | 30 s | | Offline queue retention | 30 days | | Supported browsers | 4 (Chrome, Firefox, Opera, Yandex) | | Functional requirements | 25 | | Non-functional requirements | 15 |

**Deliverable inventory (master-guide Â§1â€œFile Count Summaryâ€œ/â€œKey Statisticsâ€œ):** 60 source files; 36 diagrams (12 types — 5 formats); 5 markdown docs (25,725+ words); research = 27,560 lines across 12 dimensions; 320+ test cases (unit+E2E); 80% + coverage target; ~4.8 MB total. **133 files / 8 formats.**

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## B. FUNCTIONAL REQUIREMENTS (25 FRs â€” tech-spec Â§3, verbatim)

### B.1 Torrent Detection & Parsing (Â§3.1)

- **FR-001 â€” Detect Magnet Links on Any Web Page** (P0 Critical; Content Script; trigger = DOM mutation via `MutationObserver` or page load). SHALL detect all magnet URI links (`href` beginning `magnet:?`) in DOM of any page; detection MUST be dynamic across SPA navigation. **Acceptance:** magnet links in dynamically injected content (React/Vue re-renders) detected within **500 ms**; works across all `http://https:` origins; `iframe` magnet links detected when `all_frames` enabled.
- **FR-002 â€” Detect .torrent File Links** (P0 Critical; Content Script). Detect via URL path suffix `.torrent` (case-insensitive) AND `Content-Type: application/x-bittorrent` response header (when accessible via CORS or Fetch API).
- **FR-003 â€” Parse Magnet URI Parameters** (P0 Critical; Parser Module). Per **BEP-0009**: `xt` (Exact Topic, `urn:bti:` + 40-char hex infohash), `dn` (Display Name), `tr` (Tracker URL, repeatable), `xl` (Exact Length bytes, optional), `ws` (Web Seed, optional), `x.pe` (Peer Address host:port, optional). **Validation:** `xt` MUST be present + valid 40-char hex infohash; `dn` SHOULD be present, fallback "Unknown Torrent"; `tr` entries MUST be valid HTTP(S) URLs, malformed trackers logged+discarded.

- **FR-004** – Download and Parse .torrent Files (P1 High; Service Worker). Download via `fetch()`. Constraints: **max file size 10 MB** (configurable via `maxTorrentFileSize`); **timeout 30 s**; CORS policy falls back to **proxying via Boba server** when direct fetch blocked. Parse via Bencode decoder to extract `info.name`, `info.piece length`, `info.length`(or `info.files` multi-file), `announce + announce-list`.
- **FR-005** – Compute Infohash from Torrent Data (P1 High; Crypto Module). Compute **SHA-1** over Bencode-encoded `info dict` → 40-char hex infohash; **MUST** match `xt` if both available.
- **FR-006** – Deduplicate Torrents by Infohash (P1 High; State Manager). In-memory `Map<string, TorrentInfo>` keyed by **lowercase infohash**. On duplicate (same infohash, different page): preserve most descriptive `displayName` (longest string), union of all tracker URLs, earliest `detectedAt`.

## B.2 Integration and Transmission (Â§3.2)

- **FR-007** – Send Magnet Links to qBitTorrent via WebUI API (P0 Critical; API Client). Transmit to `qBitTorrent /api/v2/torrents/add`; support `urls` param (newline-separated magnets), optional `category/tags/savepath/rename`, automatic cookie-based session management.
- **FR-008** – Upload .torrent Files to qBitTorrent (P1 High; API Client). Upload via `multipart/form-data` POST to `/api/v2/torrents/add` using `torrents` file field. File blobs kept in memory only for upload duration.
- **FR-009** – Support Tab Group Batch Operations (P1 High; Tab Manager). Enumerate all tabs in a selected Chrome tab group; send `SCAN_REQUEST` to each tab's content script; aggregate `SCAN_RESPONSE`; dedup across tabs; transmit unified list as batch.
- **FR-010** – Enumerate Tab Groups and Extract URLs (P1 High; Tab Manager). Using `chrome.tabGroups.query()` + `chrome.tabs.query()`: list all tab groups in current window; display group titles + colors in popup; allow selecting one or more groups for batch.

## B.3 Service Discovery and Authentication (Â§3.3)

- **FR-011** – Auto-Discover Boba Services on Local Network (P2 Medium; Discovery Module). When no server configured: probe `https://boba.local:8443/health`, `https://boba.local:8080/health`, `https://boba:8443/health`; `fetch()` with `mode: 'no-cors' + 5 s` timeout; parse version, validate min supported `>=1.0.0`; present ranked list (fastest first); persist selection to encrypted storage. (**NOTE** – Boba flag: real Boba ports are 7186/7187/7189, NOT 8443/8080. See Â§J.)
- **FR-012** – Support Multiple Authentication Methods (P1 High; Auth Module). Methods: **Cookie-based** (auto `SID` cookie from qBitTorrent login – default for qBT direct); **API Key** (`X-API-Key` header for Boba server – default for Boba mode); **Basic Auth** (`Authorization: Basic <base64>` – optional); **Custom Header** (user-defined name+value – optional).
- **FR-013** – Encrypt Stored Credentials (P0 Critical; Crypto Module). All sensitive data to `chrome.storage.local` encrypted **AES-256-GCM** (Web Crypto API): 256-bit master key derived from per-installation random salt via **PBKDF2**; encryption key stored in `chrome.storage.session` (in-memory only, cleared on browser restart); each credential gets a unique **96-bit IV**; **128-bit auth tag** prepended to ciphertext.

## B.4 Offline and Resilience (Â§3.4)

- **FR-014** "Offline Queue with Retry (P1 High; Queue Manager). Persistent FIFO queue for failed sends: **max queue size 1,000** (configurable); retry = **exponential backoff starting 5 s, capped at 5 min; max retry attempts 5** (configurable); persisted to `chrome.storage.local`, survives restarts; items past max retries " " "Dead Letter" sub-store for manual review.
- **FR-015** "Real-time Download Progress via Badge (P2 Medium; Badge Manager). Toolbar badge displays numeric count of queued torrents; colors: green(idle), blue(sending), orange(queued/retrying), red(error); updates within **200 ms** of state change.

## B.5 User Interface (Â§3.5)

- **FR-016** "Context Menu Integration (P1 High; UI Controller). Via `chrome.contextMenus`: `Link(magnet)" "Send Magnet to Boba`; `Link(.torrent)" "Download .torrent to Boba`; `Page"` "Scan Page for Torrents"; `Tab Group"` "Send Tab Group to Boba".
- **FR-017** "Keyboard Shortcuts (P2 Medium; Command Handler). Via `chrome.commands`: `send-current-page = Ctrl+Shift+B`(`Cmd+Shift+B` mac); `open-popup = Ctrl+Shift+L`(`Cmd+Shift+L`); `send-tab-group = Ctrl+Shift+G`(`Cmd+Shift+G`). All user-customizable via `chrome://extensions/shortcuts`. (NOTE: user-guide Â§2.7 adds a 4th: "Scan page (without sending)" = `Ctrl+Shift+S`/`Cmd+Shift+S` " see Â§D.13 ambiguity.)
- **FR-018** "Cross-Browser Compatibility (P0 Critical; Build System). Single codebase targets: Chrome 88+ (Blink, primary dev target); Firefox 109+ (Gecko, polyfill for `chrome.*`); Opera 74+ (Blink, Chromium-compat); Yandex 21+ (Blink, Chromium-compat). Browser-specific code isolated behind `BrowserAdapter` abstraction.
- **FR-019** "Internationalization Support (P2 Medium; I18n Module). Via `_locales/{lang}/messages.json`: en(default), zh\_CN, zh\_TW, es, fr, de, ja, ko. All user-facing strings externalized; no hardcoded UI text.
- **FR-020** "Accessibility Compliance (WCAG 2.1 AA) (P1 High; UI Layer). Contrast 4.5:1 normal text / 3:1 large; all interactive elements keyboard-navigable (Tab order, Enter/Space); ARIA labels on icon-only buttons; screen-reader announcements via `aria-live`; visible focus indicators.
- **FR-021** "Dark/Light Theme (P2 Medium; UI Layer). Respect `prefers-color-scheme` + manual override toggle in Options; CSS custom properties for all theme-aware colors.
- **FR-022** "Notifications for Download Events (P2 Medium; Notification Manager). Via `chrome.notifications`: `Send success"` "Torrent Sent"; `Send failed"` "Send Failed" (click to retry); `Batch complete"` "Batch Complete" (`{count}` sent, `{failed}` failed); `Queue retry"` "Retrying Send" (attempt `{n}`). Suppressible per-category in Options.
- **FR-023** "Options Page Configuration (P1 High; Options Page). Provides: Server URL + connection mode (Boba proxy / qBitTorrent direct); auth credentials (masked input + test-connection button); default category/tags/download path; notification prefs; queue settings (max size, retry count, backoff strategy); keyboard shortcut display (non-editable, links to browser shortcuts page); theme (system/light/dark); reset to defaults with confirmation.
- **FR-024** "Health Check and Connection Status (P1 High; Health Monitor). Persistent status indicator: poll interval **30 s** (configurable); endpoint `/api/v2/app/version` (qBitTorrent) or `/health` (Boba); states `connected/connecting/disconnected/error`; visual indicator in popup header; tooltip shows last successful ping time.

- **FR-025** “Rate Limiting for API Calls (P1 High; API Client). Client-side: default **10 req/s burst, 60 req/min sustained**; queue processing = max 1 concurrent API call + 500 ms inter-call delay; respects HTTP 429 Retry-After; rate-limit state in-memory (non-persistent).

## C. NON-FUNCTIONAL REQUIREMENTS (15 NFRs “tech-spec §4)

### C.1 Performance (§4.1)

| ID      | Requirement                         | Target   | Measurement                          |
|---------|-------------------------------------|----------|--------------------------------------|
| NFR-001 | Content script initialization       | ≤ 10 ms  | <code>performance.now()</code> delta |
| NFR-002 | Magnet link detection (per link)    | ≤ 5 ms   | benchmark on 1,000-link page         |
| NFR-003 | Popup render time                   | ≤ 100 ms | Lighthouse perf audit                |
| NFR-004 | Options page load time              | ≤ 200 ms | Lighthouse perf audit                |
| NFR-005 | API call round-trip (local network) | ≤ 500 ms | p95 over 100 calls                   |
| NFR-006 | Service worker cold start           | ≤ 50 ms  | Chrome DevTools Performance          |
| NFR-007 | Bundle size (compressed CRX)        | ≤ 350 KB | <code>du -h</code> on artifact       |

### C.2 Security (§4.2)

| ID      | Requirement                     | Implementation                                          |
|---------|---------------------------------|---------------------------------------------------------|
| NFR-008 | Credential encryption at rest   | AES-256-GCM via Web Crypto API                          |
| NFR-009 | No cleartext credential logging | ESLint rule + code review                               |
| NFR-010 | HTTPS-only API communication    | URL scheme validation + CSP                             |
| NFR-011 | Content Security Policy         | <code>script-src 'self'; object-src 'none'</code>       |
| NFR-012 | Minimum permission model        | <code>activeTab</code> + declared host permissions only |

### C.3 Reliability (§4.3)

| ID      | Requirement                | Target                          |
|---------|----------------------------|---------------------------------|
| NFR-013 | Crash-free session rate    | ≥ 99.9%                         |
| NFR-014 | Offline queue durability   | 100% “all items survive restart |
| NFR-015 | API compatibility coverage | qBitTorrent 4.4.x through 5.x   |

## C.4 Scalability (Â§4.4)

- Concurrent tab group batch size: **unlimited** (memory-bound).
- Offline queue: **1,000 default, 10,000 maximum**.
- Detection: pages with 10,000+ links must not cause UI jank (> 16 ms frame time).

## C.5 Maintainability (Â§4.5)

- Coverage: **80% unit, 60% E2E** scenario.
  - All public APIs documented w/ JSDoc.
  - TypeScript strict mode; **zero** any in production.
  - **20 runtime dependencies**.
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# D. DATA MODELS & TYPE DEFINITIONS

## D.1 TorrentInfo (tech-spec Â§6.1; api-ref Â§5.2)

infohash (40-char lowercase hex), displayName, magnetUri, source ('magnet-link' | 'torrent-file'), trackers: string[], totalSize?: number, webSeeds?: string[], pageUrl, detectedAt (Unix ms), fileBlob?: Blob (torrent uploads only).

## D.2 ServerConfig (Â§6.2; api-ref Â§5.2)

mode ('boba' | 'qbittorrent-direct'), baseUrl, authMethod ('cookie' | 'api-key' | 'basic' | 'custom-header'), credentials: EncryptedCredentials, defaultCategory?, defaultTags?: string[], defaultSavePath?, timeout, healthCheckInterval.

## D.3 ExtensionConfig (Â§6.3; api-ref Â§5.2)

version, server: ServerConfig, ui {theme: 'system' | 'light' | 'dark', badgeEnabled, notifications {sendSuccess, sendFailed, batchComplete, queueRetry}}, queue {maxSize, maxRetries, retryBaseDelayMs, retryMaxDelayMs}, detection {scanDynamically, maxTorrentFileSize, enableTabGroupScan}, rateLimit {requestsPerSecond, requestsPerMinute, interCallDelayMs}.

## D.4 QueueItem (Â§6.4; api-ref Â§5.3)

id, torrent: TorrentInfo, status ('pending' | 'retrying' | 'failed' | 'dead-letter'), attemptCount, nextRetryAt (Unix ms), lastError?, lastHttpStatus?, queuedAt.

## D.5 EncryptedCredentials (api-ref Â§5.2)

salt (base64, 16 bytes), ciphertext (base64 = IV 12 bytes + ciphertext + authTag 16 bytes), encryptedAt.

## D.6 Other types (api-ref Â§5)

Scalar types: Infohash (/^[a-f0-9]{40}\$/), MagnetUri (/^magnet:\?xt=urn:btih:[a-fA-F0-9]{40}/), Priority ('low'|'normal'|'high'|'critical'), ConnectionState, AuthMethod, ConnectionMode, ThemePreference, QueueStatus, TorrentSource. Also ApiResponse<T>, ApiError, ConnectionTestResult, QueueSnapshot, RateLimitBucket, DeepPartial<T>, MessageHandler<>. Full qBitTorrent types: QBitTorrentInfo (40+ fields), TorrentState (18-value union: error, missingFiles, uploading, pausedUP, queuedUP, stalledUP, checkingUP, forcedUP, allocating, downloading, metaDL, pausedDL, queuedDL, stalledDL, checkingDL, forcedDL, checkingResumeData, moving), ServerState, MainDataResponse.

## D.7 SQL Schema (implementation-plan Â§â€œSQL Schema Definitionsâ€)

Models chrome.storage.local as SQL-shaped tables: extension\_config(key PK, value, updated\_at); discovered\_torrents(id=infoHash PK, page\_url, magnet\_uri, torrent\_url, name, trackers JSON, size\_bytes, source\_type CHECK('magnet-link'|'torrent-file'|'torrent-url'), discovered\_at, sent\_to\_boba, boba\_status CHECK('pending'|'queued'|'added'|'error')); download\_queue(id AUTOINC, info\_hash, magnet\_uri, torrent\_data BLOB base64, name, category, save\_path, added\_at, retry\_count, last\_error, status CHECK('pending'|'retrying'|'failed'|'completed')); server\_config(id=1, base\_url, auth\_method CHECK('none'|'cookie'|'api\_key'|'basic'), api\_key, username, password\_encrypted, is\_reachable, last\_check, qbittorrent\_version, boba\_version); send\_history(id AUTOINC, info\_hash, name, sent\_at, success, error\_message, server\_url); optional search\_cache(id AUTOINC, query, category, results\_json, cached\_at, expires\_at). **NOTE:** schema vocabularies disagree with TS types (source\_type adds 'torrent-url'; status uses 'completed' not 'dead-letter'; auth\_method adds 'none', drops 'custom-header') â€” see Â§K ambiguities.

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## E. APIs / ENDPOINTS

### E.1 Internal message-passing protocol (tech-spec Â§7.1; api-ref Â§2)

Uniform envelope ExtensionMessage<T> { type: MessageType, requestId?, payload: T, sender?{tabId?,frameId?,url?} }. **MessageType union (23 values):** SCAN\_REQUEST, SCAN\_RESPONSE, TORRENT\_DETECTED, SEND\_TORRENT, SEND\_BATCH, GET\_STATUS, STATUS\_RESPONSE, GET\_QUEUE, QUEUE\_RESPONSE, RETRY\_ITEM, REMOVE\_ITEM, CLEAR\_QUEUE, GET\_CONFIG, CONFIG\_RESPONSE, UPDATE\_CONFIG, CONFIG\_UPDATED, GET\_HEALTH, HEALTH\_RESPONSE, DISCOVER\_SERVERS, DISCOVERY\_RESPONSE, TEST\_CONNECTION, CONNECTION\_TEST\_RESULT.

Per-message payloads (api-ref Â§2.2â€“2.10): - SCAN\_REQUEST (SWâ†’CS via chrome.tabs.sendMessage): {requestId, deepScan, scope?}. - SCAN\_RESPONSE (CSâ†’SW): {requestId, torrents: TorrentInfo[],

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error?, meta:{linksScanned, magnetsFound, torrentFilesFound,
scanDurationMs}}.- TORRENT_DETECTED (CSâ†’SW via
chrome.runtime.sendMessage):{torrents, pageUrl, timestamp}.-
SEND_TORRENT (Popup/CtxMenuâ†’SW): {torrent, priority?:
'normal'|'high', category?, tags?}.- SEND_BATCH (TabMgrâ†’SW):
{torrents, category?, tabGroupId?};response {accepted, rejected,
failed, results:[{infohash,
status:'queued'|"sent"|"duplicate"|"failed', error?}]}.-
GET_STATUSâ†’STATUS_RESPONSE:{connection, serverUrl,
serverVersion?, queueSize, pendingCount, deadLetterCount,
lastHealthCheck?, extensionVersion}.- GET_QUEUE:{filter?, limit?
(def 100), offset?}â†’{items, total, filtered}.- RETRY_ITEM
{itemId}; REMOVE_ITEM {itemId}; CLEAR_QUEUE {status?}.- GET_CONFIG
(empty)â†’full ExtensionConfig; UPDATE_CONFIG {config:
DeepPartial<ExtensionConfig>} (SW validates, merges, persists encrypted, broadcasts
CONFIG_UPDATED).- DISCOVER_SERVERS {timeout?(def 5000), targets?}â†’
DISCOVERY_RESPONSE {servers:
[{url,name,version,responseTimeMs,healthy}], targetsProbed,
durationMs}.

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## E.2 Boba API integration (tech-spec Â§7.2; api-ref Â§3) â€” base <https://host}:{port}/api/v1>

- POST /api/v1/auth/token â€” body {apiKey} â†’ {token(JWT), expiresAt, type:"Bearer"}; 401 â†’ {error:"E\_AUTH\_INVALID\_KEY", message}.
- POST /api/v1/auth/refresh â€” Authorization: Bearer {jwt} â†’ {token, expiresAt}.
- GET /api/v1/health â€” no auth â†’ {status:"healthy", version, uptime, extensions:{qbittorrent, search, metadata}}.
- GET /api/v1/torrents/search?q=&category=&limit=(def 50,max 100)&offset= â€” header X-API-Key â†’ {results:[{infohash, displayName, magnetUri, size, seeders, leechers, source, category, uploadedAt}], total, query, categories}.
- POST /api/v1/torrents/download â€” X-API-Key + JSON {magnetUri?| infohash?, category?, tags?, savePath?, rename?, paused?, skipChecking?, sequentialDownload?, firstLastPiecePrio?} â†’ 202 {jobId, status:"queued", infohash, estimatedStart}; 409 â†’ {error:"E\_DUPLICATE", message, existingInfohash}.
- GET /api/v1/events/download-progress â€” X-API-Key, Accept: text/event-stream (SSE). Event types: progress {infohash, progress(0â€”1), speed, eta, downloaded, total, numSeeds, numLeechs, state}, completed {infohash, completionTime, ratio}, error {infohash, error, code}, stopped {infohash, reason}.

## E.3 qBitTorrent WebUI API v2 (api-ref Â§4) â€” base e.g. <https://localhost:8080>

**ALL endpoints require Referer header matching qBitTorrent origin** (else rejected); session via SID cookie. - POST /api/v2/auth/login â€” form username,password; 200 body Ok. SID; 403 invalid; 429 too many attempts. - POST /api/v2/auth/logout. - POST /api/v2/torrents/add â€” params: urls(newline-sep magnets/HTTP) *OR* torrents(*multipart file*) [â€” required], savepath, cookie, category, tags(comma-sep), skip\_checking, paused, root\_folder, rename, upLimit, dlLimit,



ratioLimit, seedingTimeLimit, autoTMM, sequentialDownload, firstLastPiecePrio. Responses: 200 ok, 400 missing urls/torrents, 403 not auth, 415 unsupported media type, 500. - GET /api/v2/torrents/info?filter=&category=&tag=&sort=&reverse=&limit=&offset=&hashes= (filter values: all, downloading, seeding, completed, paused, active, inactive, resumed, stalled, stalled\_uploading, stalled\_downloading, errored; hashes pipe-separated). - POST /api/v2/torrents/delete â€” hashes(pipe-sep or all), deleteFiles. - POST /api/v2/torrents/pause â€” hashes. - POST /api/v2/torrents/start â€” hashes. - POST /api/v2/torrents/recheck â€” hashes. - POST /api/v2/torrents/reannounce â€” hashes. - GET /api/v2/torrents/categories. - POST /api/v2/torrents/createCategory â€” category, savePath?. - GET /api/v2/torrents/tags. - POST /api/v2/torrents/addTags â€” hashes, tags. - POST /api/v2/torrents/removeTags (listed in Â§4.7 summary). - GET /api/v2/sync/maindata?rid= (0 = initial, then last rid). - GET /api/v2/app/version (plain text e.g.Â v4.6.4). - GET /api/v2/app/webapiVersion (plain text e.g.Â 2.9.3). - GET /api/v2/app/preferences / POST /api/v2/app/setPreferences (json=). - GET /api/v2/app/buildInfo (qt/libtorrent/boost/openssl/zlib). - GET /api/v2/transfer/info.

## E.4 Base URLs (api-ref Â§â€œBase URLsâ€)

| Env             | Boba                        | qBitTorrent                  |
|-----------------|-----------------------------|------------------------------|
| Local (default) | https://<br>boba.local:8443 | https://<br>localhost:8080   |
| Docker          | https://boba:8443           | https://<br>qbittorrent:8080 |
| Custom          | user-configured             | user-configured              |

## F. ERROR HANDLING

### F.1 Tech-spec Â§9.1 categories

E\_NETWORK (backoff, silent/badge), E\_TIMEOUT (immediate retry —2 then backoff, silent), E\_AUTH (no retry, immediate notification), E\_RATE\_LIMIT/429 (honor Retry-After then backoff, silent), E\_VALIDATION (no retry, immediate notification), E\_SERVER/5xx (backoff+jitter, badge+optional notif), E\_STORAGE (no retry, immediate notification). ExtensionError {code, message, timestamp, context?, recoverable}.

### F.2 api-ref Â§6.1 full error-code catalogue (22 codes)

E\_UNKNOWN, E\_NETWORK, E\_TIMEOUT(408), E\_DNS, E\_CONN\_REFUSED, E\_AUTH(401), E\_AUTH\_EXPIRED(401), E\_RATE\_LIMIT(429), E\_CLIENT\_RATE\_LIMIT, E\_VALIDATION(400), E\_SERVER(500), E\_SERVICE\_UNAVAILABLE(503), E\_TORRENT\_INVALID(400), E\_TORRENT\_DUPLICATE(409), E\_STORAGE\_FULL, E\_STORAGE\_CORRUPT, E\_CRYPTO, E\_PERMISSION\_DENIED(403), E\_TAB\_ACCESS, E\_CORS, E\_FILE\_TOO\_LARGE(413), E\_UNSUPPORTED\_BROWSER. Boba error body: {error, message, retryAfter?, details{limit, window, remaining}}.

## F.3 Retry strategies (api-ref §6.4)

Network: backoff, 5 retries (5s+10s+20s+40s+80 s). Timeout: backoff, 3 (2s+4s+8 s). Rate-limit/429: honor Retry-After, 10. Auth: no retry. 5xx: backoff+jitter, 5 (5s+10s+20s+40s+80 s). Validation: no retry.

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## G. CONFIG / SETTINGS (Options page “ user-guide §3.1, installation-guide §5)

**Server Settings:** Connection Mode (Boba/qBT-direct), Server URL, Authentication (API Key / Username+Password / Basic / Custom Header), Test Connection, Connection Timeout (default **30 s**), Health Check Interval (default **30 s**). **Download Preferences:** Default Category (“”), Default Tags (“”; e.g. browser, auto), Default Save Path (“”; server-side path), Pause After Add (Off), Skip Hash Check (Off), Sequential Download (Off), First/Last Piece Priority (Off). **Queue Settings:** Max Queue Size (**1,000**), Max Retries (**5**), Base Retry Delay (**5 s**), Max Retry Delay (**300 s**). **Notification Settings:** Send Success (Enabled), Send Failed (Enabled), Batch Complete (Enabled), Queue Retry (Disabled). **Detection Settings:** Dynamic Scanning (On), Highlight on Page (Off), Max File Size MB (**10**), Tab Group Scanning (On). **UI Settings:** Theme (System/Light/Dark; default System), Show Badge (On), Badge Color (Default; per-state custom). **Security Settings:** Require Password on Startup (Off), Auto-Lock after N min inactivity (**30**), HTTPS Only (On), Certificate Pinning (Off). “Reset to Defaults” with confirmation; “Forgot Password” “no recovery, must reset extension.”

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## H. USER-FACING FLOWS & USE CASES

### H.1 Core workflows (tech-spec §2.2)

- **W1 Single magnet detect+forward:** content script observes DOM mutations “ match regex ^magnet:\?xt=urn:btih:[a-fA-F0-9]{40} “ parse to TorrentInfo “ chrome.runtime.sendMessage() “ SW enqueues or sends per connectivity “ target API initiates download.
- **W2 Tab group batch ingestion:** user triggers “Send Tab Group to Boba” “ SW enumerates group tabs “ each gets SCAN\_REQUEST “ aggregate, dedup by infohash, transmit batch.
- **W3 Auto-discovery:** on first install / empty config, send mDNS/Bonjour-like probes to well-known local addresses “ responsive Boba instances reply endpoint+version “ user picks from ranked list “ persists to encrypted storage.

### H.2 Send methods (user-guide §2.2)

1. Context menu right-click “Send Magnet to Boba”; (2) Popup “ check torrents “Send Selected”; (3) keyboard Ctrl+Shift+B; (4) Popup “Send All”.

### H.3 .torrent file send (user-guide §2.3)

Context menu “Download .torrent to Boba” downloads (≈10 MB), parses, sends. CORS fallback: Boba mode auto-proxies; qBT-direct shows “CORS blocked” “ user copies link to qBT manually.

## H.4 Tab groups batch (user-guide Â§2.4)

Methods: Popup "Tab Groups" tab "Send" next to group; right-click tab "Send Tab Group to Boba"; Ctrl+Shift+G. Result: scans every tab, collects, removes cross-tab dups, batch sends, summary notif ("Batch Complete" 12 sent, 0 failed).

## H.5 Queue management (user-guide Â§2.5)

Popup "Queue" tab "columns Name/Status/Attempts/Next Retry/Actions. Actions: Retry now (circular arrow), Remove (X, no recovery), Retry All Failed, Clear Queue (confirm). Behavior: auto-retry exponential backoff (5/10/20/40/80 s); max 5 'Dead Letter'; dead-letter requires manual action; queue persists across restarts/reboots.

## H.6 Download progress (user-guide Â§2.6)

Toolbar badge (number = queued/sending count; colors green/blue/orange/red/gray). Popup "Downloads" tab (SSE, Boba mode only): Name/Progress bar+%/Speed/ETA/Status. qBT-direct = no live progress in popup (check qBT WebUI).

## H.7 First-time setup wizard (user-guide Â§1.4; installation-guide Â§4)

Step 1 choose Connection Mode (Boba Server recommended / qBitTorrent Direct); Step 2 (Boba) server discovery "Discover Servers" scans boba.local:8443,boba:8443, ranked list, "Connect"; manual fallback (URL + API key + Test Connection); Step 3 (qBT-direct) URL + username/password + Test Connection + optional default category/path; Step 4 "Finish Setup".

## H.8 Connection status icon states (user-guide Â§1.5)

Green dot=connected; Blue spinner=connecting/sending; Orange dot=queued pending; Red dot=connection error; Gray dot=not configured.

## H.9 Detection visuals (user-guide Â§2.1)

Default = no page modification. Optional "Highlight torrents on page": magnet links green underline, .torrent links blue underline, hover tooltip with name+size.

## H.10 Power-user workflows (user-guide Â§5.4)

Nightly batch download (accumulate pages in "To Download" group "Ctrl+Shift+G at night); category-based org (set categories + qBT Automatic Torrent Management auto-move on completion); monitoring large batches (enable all notifications + watch).

## H.11 Verification flow (installation-guide Â§6)

Test on known torrent site (Ubuntu downloads), verify detection (badge count, popup list name/size/source), verify send ("Send Selected" "notif), check qBT for download, queue test (disconnect"send"reconnect"item sent), settings-persist test (restart"Options retained).

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## I. EDGE CASES (explicitly documented)

- Magnet without dn â†’ display â€œUnknown Torrentâ€¢, name resolved by qBT from metadata (FR-003; FAQ Q16).
  - Duplicate torrent (same infohash) â†’ silently skipped, â€œduplicateâ€¢ status; dedup merges longest displayName + union trackers + earliest timestamp (FR-006; FAQ Q17; api-ref dedup).
  - Malformed tracker URLs â†’ logged + discarded (FR-003).
  - .torrent > 10 MB â†’ not downloaded; E\_FILE\_TOO\_LARGE(413) â†’ â€œUse magnet link insteadâ€¢ (FR-004; api-ref Â§6.1).
  - CORS-blocked .torrent download â†’ Boba proxies (Boba mode); qBT-direct shows â€œCORS blockedâ€¢, manual copy (FR-004; user-guide Â§2.3; E\_CORS â†’ â€œUse Boba proxy modeâ€¢).
  - Dynamically-loaded links (SPA/infinite scroll) â†’ MutationObserver, debounced 250 ms (tech-spec Â§10.2) / 500 ms (impl-plan Â§2.3.5); â€œwait a few secondsâ€¢ (user-guide Â§4.2).
  - Links inside iframes â†’ detected only when all\_frames enabled; may be inaccessible (FR-001; user-guide Â§4.2).
  - JS click-handler â€œlinksâ€¢ (not real <a>) â†’ not detected; manual copy (user-guide Â§4.2).
  - Non-standard/encoded magnet formats â†’ not detected; manual copy (user-guide Â§4.2).
  - Site requires login / shows links only to logged-in users (user-guide Â§4.4).
  - Strict-CSP pages â†’ extension respects CSP, will NOT inject scripts on such pages (user-guide Â§4.2/Â§4.4; tech-spec threat model).
  - Forgotten encryption password â†’ no recovery; reset extension + reconfigure (user-guide Â§3.5; FAQ Q26; risk table).
  - Browser restart without password protection â†’ session encryption key lost â†’ â€œCannot decrypt credentialsâ€¢; re-enter credentials (dev-guide Â§9.5).
  - Storage quota exceeded â†’ E\_STORAGE\_FULL; Firefox stricter limits, clear old queue items (api-ref Â§6.1; installation-guide Â§9.8).
  - Pages with 10,000+ links â†’ must not jank (>16 ms frame); requestIdleCallback for >500 links; pagination for large link sets (NFR Â§4.4; tech-spec Â§10.2; risk table).
  - qBitTorrent rejects duplicate â†’ may already be downloading/completed (user-guide Â§4.3).
  - Private tracker torrents â†’ work if magnet/.torrent accessible; may need passkeys in tracker URL (FAQ Q15).
  - MV3 service worker termination â†’ chrome.alarms keep-alive (dev-guide Â§6.1; risk table; impl-plan Â§3.2.3); Firefox suspends SW more aggressively (installation-guide Â§9.8).
  - Self-signed certs â†’ may need browser exception / system trust install; â€œHTTPS Onlyâ€¢ can be temporarily disabled for testing (installation-guide Â§9.6; user-guide Â§4.1).
  - macOS \_\_MACOSX metadata folder in ZIP â†’ load error, must delete (installation-guide Â§9.5).
  - Mobile browsers unsupported (no full WebExtensions API / SW / tab groups) (FAQ Q4).
  - Safari only â€œPartialâ€¢ MV3, not tested (api-ref Â§8.2).
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## J. BACKEND INTEGRATION POINTS

### J.1 Two operating modes

- **Boba Server mode** (recommended): extension `Bob REST /api/v1/*` Boba proxies/orchestrates to qBitTorrent. Adds: search aggregation, metadata enrichment, SSE progress streaming, auto-discovery, CORS proxy for .torrent downloads. Auth = API Key (X-API-Key) or JWT (Authorization: Bearer).
- **qBitTorrent Direct mode**: extension `qBitTorrent WebUI /api/v2/*` directly. Auth = cookie (SID) via `/api/v2/auth/login`, or Basic Auth. Referer header mandatory. No live progress in popup.

### J.2 Spec-documented endpoints the extension calls

- Send magnet/.torrent: Boba `POST /api/v1/torrents/download` OR qBT `POST /api/v2/torrents/add`.
- Health: Boba `GET /api/v1/health` (or `/health`); qBT `GET /api/v2/app/version`.
- Progress: Boba SSE `GET /api/v1/events/download-progress`; qBT polling `GET /api/v2/sync/maindata` + `GET /api/v2/torrents/info`.
- Search: Boba `GET /api/v1/torrents/search`.
- Categories/tags: qBT `GET/POST /api/v2/torrents/{categories,createCategory,tags,addTags,removeTags}`.

### J.3 Mapping onto REAL Boba services (per task brief “flagged for conductor)

The spec assumes generic Boba ports **8443/8080**, but the real Boba repo runs: - **download-proxy on port 7186** qBitTorrent WebUI (port 7185 internal). The qBT-direct `/api/v2/*` calls SHOULD target **7186** (the proxy), which already injects auth cookies for private trackers; WebUI creds are hardcoded `admin/admin`. - **merge-search service on port 7187** (FastAPI or Go/Gin) maps to the spec’s Boba search/dashboard role (`/api/v1/torrents/search`, SSE progress, dashboard at `http://localhost:7187/`). The spec’s “Boba Server” REST surface is best mapped here. - **boba-jackett on port 7189** (Go) owns Jackett credentials/indexer overrides/autoconfig; relevant to the “search aggregation” feature. - **CRITICAL FINDING “implementation-plan §4.1.4 ALREADY cites the real ports: “Auto-discovery”** Scan localhost ports **7187, 7189, 8080**. This is the strongest signal the extension’s backend discovery should target Boba’s real 7187/7189 (and the proxy 7186), NOT the spec’s fictional 8443. - Auto-discovery probe targets in FR-011/installation-guide (`boba.local:8443`, `boba:8443`, `localhost:8443`, `:8080`) must be **re-pointed to 7186/7187/7189** for the Boba implementation.

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## K. BOBA-CONSTRAINT FLAGS (task-mandated callouts)

### K.1 “CI/CD” Boba FORBIDS all CI/CD; the reference relies heavily on GitHub Actions

The reference deliverable **assumes GitHub Actions CI/CD** in multiple places “ALL of these violate Boba’s permanent Hard-Stop “NO CI/CD pipelines” rule and MUST be dropped/replaced with a manual `.ci.sh`-style script: - master-guide / README §src tree:

.github/workflows/ci.yml (lint, test, build, E2E) and .github/workflows/release.yml (release to all stores). - README “Verification Checklist”: “CI/CD pipelines (GitHub Actions) ...”. - tech-spec §12.2 “CI/CD Pipeline” mermaid: Developer Push → Lint → Unit → Build → E2E → Coverage Gate → (main) Deploy to Chrome Web Store / AMO / GitHub Release. - dev-guide §1.1: Git workflow GitHub Flow; §7.5 release process auto-submits to stores; coverage thresholds “enforced in CI”. - impl-plan §8.1.3 ci.yml (lint/test/build) + §8.1.4 release.yml (publish to stores). - **Action for conductor:** strip every .github/workflows/\*.yml; convert lint/test/build/E2E/coverage into a manual script (Boba’s ./ci.sh analogue). No auto-trigger on push/PR. Husky pre-commit hook (impl-plan §1.1.4) is ALSO a git-hook → Boba forbids git hooks → drop.

## K.2 Hardcoded localhost/ports vs CONST-XII (no hardcoded client-facing URLs)

The spec hardwires localhost:8080, boba.local:8443, etc. Boba CONST-XII forbids hardcoded localhost/127.0.0.1 for client-facing URLs “but for a browser extension the server URL is user-configured (Options), which satisfies the spirit. Discovery defaults must derive from config, not literals. Flag for review but lower severity.

## K.3 Boba credentials reality

- qBitTorrent WebUI creds are **hardcoded** admin/admin in Boba “the extension”’s qBT-direct auth defaults can assume this for local proxy (7186) but must not commit any real secret. Spec’s PBKDF2/AES-256-GCM credential encryption still applies for user-entered Boba/tracker keys.
- Never commit .env / tracker credentials (Boba rule) “extension build must not bundle them.

## K.4 TDD / anti-bluff

Reference targets 80% unit / 60% E2E coverage with Jest+Playwright. Boba mandates TDD (RED → GREEN) + anti-bluff (assert user-observable outcomes: real HTTP to 7186/7187, real DOM text, real container state) “the reference”’s coverage targets are a floor; tests must drive the real proxy/merge stack, not mocks, for non-unit layers.

## K.5 Tooling alignment notes

- Reference stack: WXT 0.17, Vite 5, TS 5, Jest 29, Playwright 1.40, ESLint 8 (impl-plan says ESLint 9 flat config), Prettier 3. Boba’s existing frontend is **Angular 21 + Vitest** “the extension is a *separate* TS/WXT codebase, so Jest-vs-Vitest is a deliberate divergence to confirm with conductor (could standardize on Vitest to match Boba).
  - Runtime deps (20): bencode-js ^3.0.0, lz-string ^1.5.0, webextension-polyfill ^0.10.0. Dev deps: @types/chrome ^0.0.260, @types/jest, ts-jest, jest-environment-jsdom, @playwright/test.
  - LICENSE: impl-plan §1.1.1 says **Apache 2.0**; master-guide §1.1.1-adjacent unclear; FAQ Q2 says “free and open-source under its license terms” “confirm with conductor.
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## L. DEPENDENCIES & TECH STACK (tech-spec Â§11; dev-guide Â§1.3)

| Layer     | Tech                        | Version                             |
|-----------|-----------------------------|-------------------------------------|
| Language  | TypeScript                  | â€¢ 5.0                             |
| Framework | WXT                         | â€¢ 0.17                            |
| Bundler   | Vite (via WXT)              | â€¢ 5.0                             |
| Unit test | Jest                        | â€¢ 29.0                            |
| E2E test  | Playwright                  | â€¢ 1.40                            |
| Lint      | ESLint                      | â€¢ 8.0 (impl-plan: 9 flat config)  |
| Format    | Prettier                    | â€¢ 3.0                             |
| Crypto    | Web Crypto API              | native (AES-256-GCM, SHA-1, PBKDF2) |
| Storage   | Extension Storage API (MV3) | chrome.storage.local/session        |
| UI        | Vanilla TS + CSS            | no UI framework                     |

**Runtime deps:** bencode-js ^3.0.0, lz-string ^1.5.0, webextension-polyfill ^0.10.0. **External deps (tech-spec Â§13.2):** Node.js 18 LTS (MIT), qBitTorrent WebUI 4.4.0 (GPL-2.0+), Boba Server 1.0.0 (proprietary), Chrome/Chromium 88, Firefox 109 (MPL-2.0). **Dev prereqs (dev-guide Â§1.1):** Node.js 18 LTS (rec 20 LTS), npm 9.x (rec 10.x), Git 2.40+, a supported browser. IDE: VS Code (settings + recommended extensions: Prettier, ESLint, Tailwind CSS IntelliSense, Coverage Gutters, Jest).

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## M. BUILD / RELEASE STEPS

### M.1 Build commands (dev-guide Â§3)

- `npm install` (2â€“5 min).
- `npm run dev` â†’ WXT dev mode + HMR â†’ `.output/chrome-mv3-dev/`. Load unpacked per browser.
- `npm run build` â†’ production builds: `.output/{chrome-mv3-prod, firefox-mv3-prod, opera-mv3-prod, edge-mv3-prod}/` (minified, tree-shaken, source-mapped, ZIP-ready).
- Per-browser: `npx wxt build -b chrome/ -b firefox/ --browser opera`.
- Browser differences via WXT browser-specific entry points (`popup/index.firefox.ts`), `import.meta.browser` conditional compilation, `webextension-polyfill`.
- `wxt.config.ts` (dev-guide Â§3.4): `srcDir src, outDir .output, manifest perms ['activeTab', 'storage', 'contextMenus', 'notifications'], host_permissions: ['https://*/'], action.default_popup, options_ui (open_in_tab), commands`.

### M.2 Test commands (dev-guide Â§4)

- `npm test / npx jest` (`--coverage, --watch, single file, --testNamePattern`). Coverage â†’ `coverage/` (`lcov-report HTML, lcov.info, coverage-summary.json`).

- Coverage thresholds (CI-enforced): **Statements 80%, Branches 75%, Functions 80%, Lines 80%**. Per-file targets: api-client 90%, crypto-utils 95%, queue-manager 85%, magnet-scanner 80%, magnet-uri 95%.
- E2E: `npx playwright install chromium firefox; npm run test:e2e; --headed, --ui, --project=chromium, single spec.`

## M.3 Release process (dev-guide Â§7.5)

Accumulate on main â†’ release branch â†’ bump version in manifest â†’ update CHANGELOG.md â†’ full test suite â†’ Git tag â†’ build all targets â†’ GitHub Release â†’ submit Chrome Web Store + AMO + Opera Addons. SemVer (Major=breaking, Minor=new features, Patch=fixes). (All store-submission + GitHub Release auto-steps are CI/CD â†’ see Â§K.1 flag.)

## M.4 Distribution channels (tech-spec Â§12.3)

| Channel          | Format | Update    |
|------------------|--------|-----------|
| Chrome Web Store | CRX    | automatic |
| Firefox AMO      | XPI    | automatic |
| Opera Addons     | CRX    | automatic |
| GitHub Releases  | ZIP    | manual    |

## M.5 Git conventions (dev-guide Â§7)

GitHub Flow; branch prefixes feature/bugfix/hotfix/docs/refactor/test; **Conventional Commits** (feat, fix, docs, style, refactor, test, chore, perf, security; scopes popup/options/content/background/api/queue/auth/crypto/i18n/build/deps; ! + BREAKING CHANGE: footer). PR template + code-review checklist provided.

## M.6 Store submission specifics (impl-plan Â§8.2)

Chrome Web Store (screenshots, description, privacy policy, **\$5 fee**); Firefox AMO (web-extension, listing, review responses); Opera Addons (Opera build, sidebar screenshot); Yandex (Chromium-compatible package, description).

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# N. SECURITY ARCHITECTURE (tech-spec Â§8)

- **Threat model:** credential theft from disk (High â†’ AES-256-GCM + key in session-only storage); MITM (High â†’ HTTPS-only, optional cert pinning); XSS via malicious torrent site (Medium â†’ CSP, content-script isolation, no eval()); privilege escalation (Medium â†’ minimal manifest, activeTab default); extension fingerprinting (Low â†’ no unique identifiers to web pages).
  - **Encryption flow:** user enters password/API key â†’ random 16-byte salt â†’ PBKDF2(100k iterations) derive key â†’ store key in `chrome.storage.session` â†’ AES-256-GCM encrypt â†’ store `{iv+authTag+ciphertext, salt}` in `chrome.storage.local`. On startup: check session key; if exists use it, else re-derive from salt+password.
  - **CSP:** `"extension_pages": "script-src 'self'; object-src 'none'; connect-src 'self' https;";`
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## O. ARCHITECTURE & MODULE MAP

### O.1 Component layers (tech-spec §5.2)

Presentation (Popup, Options, Context Menu) → Controller (Message Router, Command Handler, Tab Manager) → Service (Parser, API Client, Queue Manager, Health) → Security (Auth Module, Crypto Module, Rate Limiter) → Storage (chrome.storage.local/session).

### O.2 SW core modules (tech-spec §5.1)

Parser, Auth, API Client, Queue Manager, Health Monitor, Tab Manager, Badge Manager, Notification Manager, Crypto Module.

### O.3 Directory layout (dev-guide §2.1) → canonical module list

src/background/ (index, message-router, init); src/content-scripts/ (index, magnet-scanner, dom-observer, torrent-parser); src/popup/ (index.html/ts, popup.css, components: torrent-list, queue-view, status-bar, tab-groups); src/options/ (index.html/ts, options.css, components: server-config, download-prefs, queue-settings, ui-settings); src/modules/ (api-client, auth-manager, badge-manager, config-manager, crypto-utils, discovery-service, health-monitor, il8n, notification-manager, queue-manager, rate-limiter, state-manager, tab-manager); src/types/ (index, messages, torrent, config, api); src/utils/ (bencode, hash, magnet-uri, storage, time, validation); tests/ (unit, e2e, mocks); \_locales/. **NOTE:** master-guide/README §src tree shows a slightly different layout (src/parser/, src/scanner/, src/api/, src/shared/, src/content/) → two competing layouts exist → see §K/§P ambiguities.

### O.3b Performance impl (tech-spec §10)

WeakMap for DOM → torrent metadata; URL .revokeObjectURL() after upload; lazy module init; MutationObserver filters for link mutations only, debounced 250 ms; requestIdleCallback for >500 links; tree-shaking; code-splitting (Options/Popup separate chunks); shared vendor chunk; LZ-string queue compression.

### O.4 Architecture decisions (dev-guide §6)

MV3 over MV2 (store requirement, ephemeral SW, lower memory; mitigate SW lifetime via chrome.alarms + chrome.storage); WXT over raw Vite (extension scaffolding, HMR, multi-browser, manifest auto-gen, file-based routing); activeTab over broad host\_permissions (privacy; optional\_permissions for proactive detection); AES-256-GCM (authenticated encryption, Web Crypto support, NIST standard; PBKDF2 100k iterations, 96-bit IV, 128-bit tag).

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## P. REFERENCE-PLAN PHASES / MILESTONES

### P.1 master plan.md → 7 Stages (research → docs)

Stage 1 Deep Research (4 agents: Boba architecture, browser-extension APIs, torrent/magnet parsing, existing-solutions+security); Stage 2 Architecture & Technical Design (→ architecture.md); Stage 3 Implementation Plan (Phase 1 → 10 listing); Stage 4 Source-code dev;

Stage 5 Diagrams (5 agents); Stage 6 Documentation (5 agents); Stage 7 Multi-format conversion (docx/pdf/html). Note: plan.md §Stage 3 lists 10 phases (Phase 1 scaffold | Phase 10 docs+release) – a DIFFERENT decomposition than implementation-plan.md’s 8 phases.

## P.2 implementation-plan.md – 8 Phases / 5-week timeline (the authoritative dev plan)

- **Phase 1 Foundation (Week 1):** 1.1 Project Setup (git init + Apache-2.0 LICENSE + README; WXT for 5 browsers; tsconfig strict + @types/chrome; ESLint 9 flat + Prettier + **Husky pre-commit**; dir structure). 1.2 Manifest V3 (base + Chrome SW + Firefox gecko ID + Opera sidebar/minimum\_opera\_version + Yandex tune://tune: URL handling + permission model storage/alarms/notifications/activeTab/host\_permissions). 1.3 Shared infra (types; logger; constants URLs/ports/regex; error classes; typed event bus).
- **Phase 2 Core Engine (Week 2):** 2.1 Magnet detect+parse (regex engine; URI decoder for dn/tr/xl; BTIH validator 40-char hex + **base32 support**; param extractor xt/dn/tr/xl/ws/xs/as/kt/mt; normalizer to {infoHash, name, trackers}; 50+ unit tests). 2.2 Torrent file parse (zero-dep Uint8Array bencode decoder; torrent parser class; SHA-1 infohash via Web Crypto; magnet generator; validator announce/info/piece length; tests). 2.3 DOM scanner (base class w/ EventTarget; anchor scanner a[href^="magnet:"], a[href\$=".torrent"]); **TreeWalker text-node scanner** for inline magnets; top-20-site CSS selector DB; MutationObserver debounced 500 ms; **Shadow DOM recursive traversal**; dedup Map by infoHash; perf budget rAF yielding 16 ms/frame).
- **Phase 3 Extension Shell (Week 2–3):** 3.1 Content Script (entry; orchestrator; highlight/overlay; message handler SCAN\_REQUEST; auto-scan document\_idle; manual scan). 3.2 Background SW (entry/lifecycle; message router; chrome.alarms keep-alive; context menu – Send to Boba – Scan Page – Open Dashboard – shortcuts; health-check loop; badge updater; notifications). 3.3 Popup (HTML/CSS theme-aware; popup.ts list/actions/state; connection status; torrent list w/ checkboxes; send + progress; batch Select All/Deselect/Invert; quick settings auto-scan/notif). 3.4 Options (HTML/CSS; options.ts save/load; Boba server config URL/port/auth/creds; **Test Connection; Auto-Discover** port scan; download prefs category/save-path/start-paused; security creds-encryption toggle + HTTPS enforcement).
- **Phase 4 API Integration (Week 3):** 4.1 Boba API Client (BobaAPIClient; auth handlers cookie/API-Key/Basic; health /app/version,/api/v2/app/version; **auto-discovery scan localhost ports 7187, 7189, 8080**; error handling backoff+classify; token-bucket rate limiter). 4.2 qBitTorrent Direct (login/SID/CSRF; add torrents; add magnets; multipart .torrent upload; monitor /torrents/info hash-filter; /sync/maindata incremental RID). 4.3 Offline Queue (OfflineQueue; chrome.storage.local persistence; backoff+jitter retry; chrome.alarms processor; dedup by infohash).
- **Phase 5 Tab Groups (Week 3):** 5.1 Enumeration (chrome.tabGroups.query({}); chrome.tabs.query({groupId})); URL aggregation; per-tab batch scan). 5.2 Context Menu (tab – Send Group to Boba – group submenu w/ per-group torrent count; scan-all+batch-send handler; progress notif).
- **Phase 6 UI/UX Polish (Week 4):** 6.1 Icons & Theming (16/32/48/128 PNG + SVG; dynamic color badges; prefers-color-scheme; animations/spinners). 6.2 i18n (\_locales/en/messages.json; English strings; chrome.i18n.getMessage() wrapper). 6.3 Accessibility (ARIA labels; keyboard nav; focus mgmt/trapping; screen-reader live regions).
- **Phase 7 Testing (Week 4–5):** 7.1 Unit (Jest+ts-jest+jsdom; chrome.\* mocks storage/tabs/runtime/alarms; magnet 50+; torrent fixtures; DOM scanner jsdom; API client mock fetch; queue persistence/retry/dedup; **80%+ line coverage**). 7.2 Integration (against real Boba docker; cookie+API-key auth; **end-to-end Detect†Parse†Send†Verify**

- download).** 7.3 E2E (Playwright config w/ extension loading; extension fixture Chromium/Chrome; popup tests; content-script detection tests; options configure+test-connection).
- **Phase 8 Build & Distribution (Week 5):** 8.1 Build pipeline (WXT prod build; cross-browser packages; `.github/workflows/ci.yml`; `release.yml` publish to stores). 8.2 Store submission (Chrome WS \$5 fee; AMO web-ext sign; Opera Addons; Yandex).

### P.3 Gantt (impl-plan)

Wk1: P1+P2start; Wk2: P2+P3start; Wk3: P3+P4+P5; Wk4: P6+P7start; Wk5: P7+P8.

### P.4 Risk assessment (impl-plan + tech-spec Â§14)

qBitTorrent API changes (High/Med â†’ version detection, adapter pattern); CORS on torrent sites (High/Med â†’ SW fetch bypasses CORS / proxy via Boba); MV3 SW termination (Med â†’ chrome.alarms keep-alive); store review rejection (Low/Med â†’ minimal permissions, guideline compliance); Yandex API differences (Low â†’ Chromium-compatible code); Chrome MV3 policy changes (Med/High â†’ MV2 fallback branch); user forgets encryption password (Med â†’ optional hint, reset/reconfigure); third-party dep vulnerability (Med/High â†’ Dependabot, lockfile pinning, SCA); large-page perf (Med â†’ requestIdleCallback, pagination).

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## Q. ACCEPTANCE CRITERIA / VERIFICATION (consolidated)

- FR-001: dynamic magnet links detected within **500 ms**; all http/https origins; iframe when all\_frames.
  - FR-003: x t valid 40-char hex required; dn fallback âœœUnknown Torrentâœœ; malformed t r discarded.
  - FR-005: computed SHA-1 infohash matches x t.
  - NFR targets (Â§C) are pass/fail acceptance gates with measurement methods.
  - Installation verification checklist (installation-guide Â§6.4): extension installed (icon), server connected (green badge), detection works (badge count >0), send works (success notif), queue works (disconnectâ†’sendâ†’reconnectâ†’sent), settings persist (restartâ†’retained).
  - Connection-test success output (installation-guide Â§4.3): Server / Mode / Status=Connected / Version / Latency / Auth=OK / API=Compatible.
  - README âœœVerification Checklistâœœ (24 items) âœœ” research, plan, source, SQL, unit (320+, 80%+), E2E (80+), CI/CD(GHA âœœ” âšš i,âœœ), spec (25 FR/15 NFR), API ref, user guide (30 FAQs), dev guide, install guide, all diagram types, multi-format docs.
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## R. VERSION COMPATIBILITY MATRICES (api-ref Â§8; installation-guide Â§7.3)

- **qBitTorrent:** 4.4.x (WebAPI 2.8.x, Compatible/legacy); 4.5.x (2.8.x, recommended min); 4.6.x (2.9.x, primary target); 5.0.x (2.10.x, compatible); 5.1.x (2.11.x, testing); <4.4.0 unsupported.
- **Browsers:** Chrome 88 min/120+ rec (full MV3, tested); Firefox 109/121+ (full w/ polyfill, tested); Opera 74/106+ (full, tested); Yandex 21/24+ (full, tested); Edge 88/120+ (full, tested); Safari 16/17+ (partial, NOT tested).

- **Boba Server:** 1.0.xâ†”i,Žext 1.0.x compatible; 1.1.x backward-compatible; 1.2.x testing; 2.0.xâ†”i,Žext 1.1.x+ planned.
- **Feature availability:** SSE progress = Boba 1.1+ only; search aggregation = Boba 1.2+ only; auto-discovery = Boba 1.0+ only; magnet send / .torrent upload / categories / tags / sequential / first-last-piece = qBT 4.4+ (tags 4.3.2+) and all Boba.

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## S. RATE LIMITING DETAIL (api-ref Â§7)

Token-bucket `TokenBucketRateLimiter(capacity, refillRate, initialTokens)`. Default buckets: per-second cap 10 refill 10/s; per-minute cap 60 refill 1/s; queue-processing cap 1 refill 2/s. Respects `X-RateLimit-Limit/Remaining/Reset + Retry-After`. Queue config: `maxConcurrent 1, interCallDelayMs 500, batchSize 10, batchIntervalMs 5000`.

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## T. OPEN QUESTIONS / AMBIGUITIES (for conductor)

1. **Real Boba port mapping (HIGH).** Spec uses fictional 8443/8080; implementation-plan Â§4.1.4 already names **7187, 7189, 8080**. Confirm: qBT-direct â†’ proxy **7186**? Boba REST/search/SSE â†’ merge service **7187**? search aggregation â†’ **boba-jackett 7189**? All FR-011 / installation-guide discovery defaults must be re-pointed. The specâ€™s separate Boba / api / v1 / \* REST surface (auth/token, search, download, SSE) may not exist on 7187 as specified â€” needs reconciliation with the actual FastAPI/Go routes.
2. **CI/CD removal (HIGH â€” Boba Hard-Stop).** Reference is GitHub-Actions-centric (ci.yml, release.yml, Husky pre-commit, coverage â€œenforced in CIâ€œâ€¢, auto store-submission). ALL must be converted to a manual `. / ci . sh`-style script + manual release; no `.github/workflows/`, no git hooks. Confirm the manual-CI shape.
3. **Jest vs Vitest (MED).** Reference mandates Jest+Playwright; Bobaâ€™s existing frontend/ is Angular 21 + **Vitest**. Standardize the extension on Vitest to match Boba, or keep Jest? Playwright E2E likely stays.
4. **Two divergent source layouts (MED).** master-guide/README src tree (src/parser, src/scanner, src/api, src/shared, src/content) vs dev-guide Â§2.1 (src/modules, src/utis, src/content-scripts). Pick one canonical layout.
5. **Two divergent phase plans (MED).** plan.md lists 10 phases; implementation-plan.md lists 8. Use the 8-phase impl-plan as authoritative?
6. **Data-model vocabulary drift (MED).** SQL schema `source_type` adds 'torrent-url'; status uses 'completed' (TS uses 'dead-letter'); `auth_method` adds 'none', omits 'custom-header'. TS `QueueStatus` and SQL status disagree. Reconcile the canonical enum sets.
7. **Magnet param set drift (LOW).** FR-003 lists `xt/dn/tr/xl/ws/x.pe`; impl-plan Â§2.1.4 lists `xt/dn/tr/xl/ws/xs/as/kt/mt` (no x.pe). Which param superset is implemented?
8. **Keyboard-shortcut count (LOW).** FR-017 defines 3 commands; user-guide Â§2.7 + impl-plan add a 4th â€œScan page (no send)â€œâ€¢ `Ctrl+Shift+S`. Confirm 3 or 4 commands; open-popup `Ctrl+Shift+L` may collide with browser defaults.
9. **host\_permissions scope (LOWâ€”MED).** `wxt.config` uses `host_permissions: [ 'https://*/' ]` while NFR-012 + dev-guide Â§6.3 push `activeTab-primary/minimal`. Reconcile (broad https â†’ store-review risk + privacy prompt).
10. **LICENSE (LOW).** impl-plan says Apache 2.0; FAQ says â€œopen-source under its license termsâ€œâ€¢. Confirm license + Boba submodule conventions.
11. **Naming (LOW).** Product is â€œBobaLinkâ€œâ€¢ throughout the reference; confirm the Boba repoâ€™s preferred extension name + icon (â€œtea bubbleâ€œâ€¢).

12. **Search/SSE feature scope (MED).** Referenceâ€™s richest features (search aggregation, SSE live progress, auto-discovery) are gated to Boba-server mode. Does the Boba implementation expose `/api/v1/torrents/search` + SSE on 7187, or must the extension call the existing merge-service routes (different shape)? Determines how much of FR-011/Â§E.2 is buildable as-specified.
  13. **Boba `/api/v1/health` vs real `/health`.** FR-024 + discovery probe `/health`; spec also uses `/api/v1/health`. Confirm the real Boba health endpoint path/port.
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